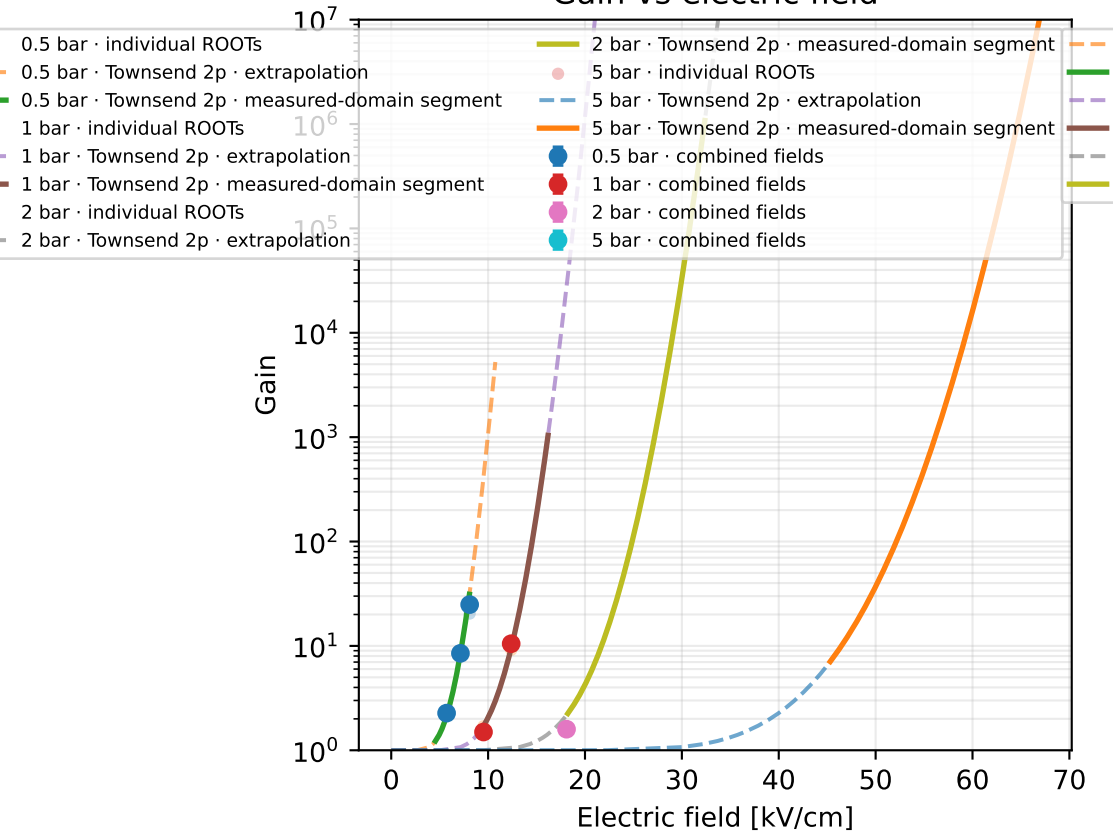
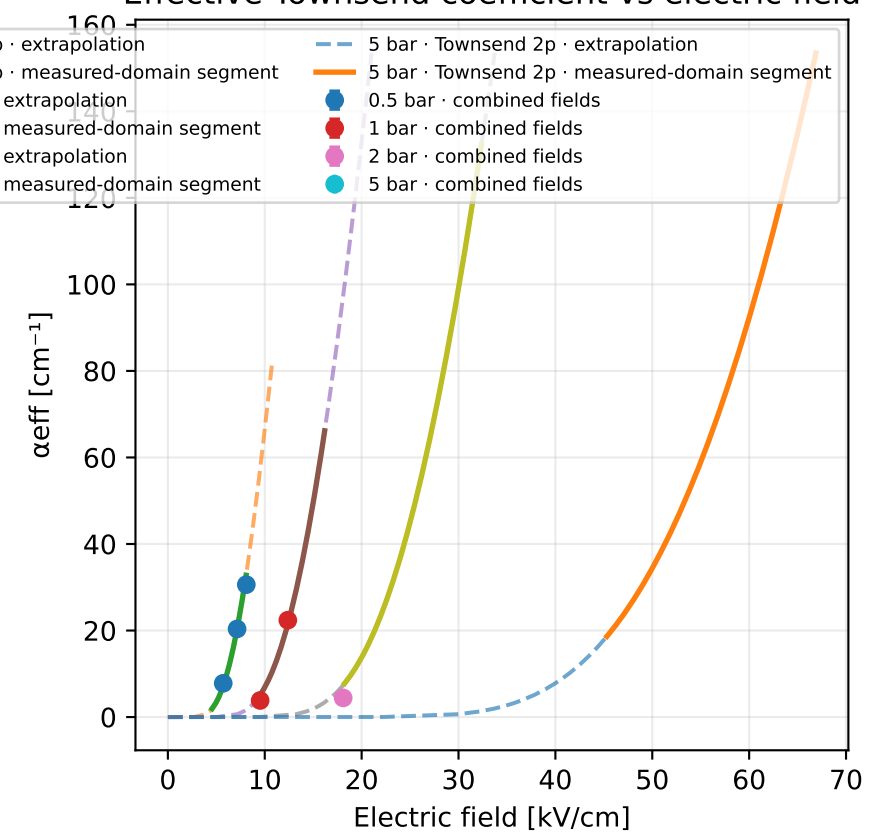
ArCF4 · Ar 99 % / CF₄ 1 % · gap 1.05 mm

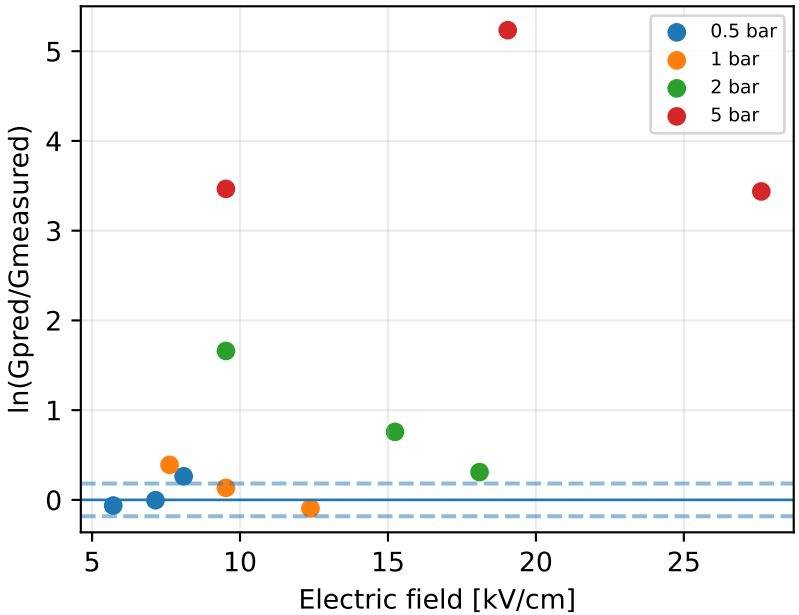
Gain vs electric field



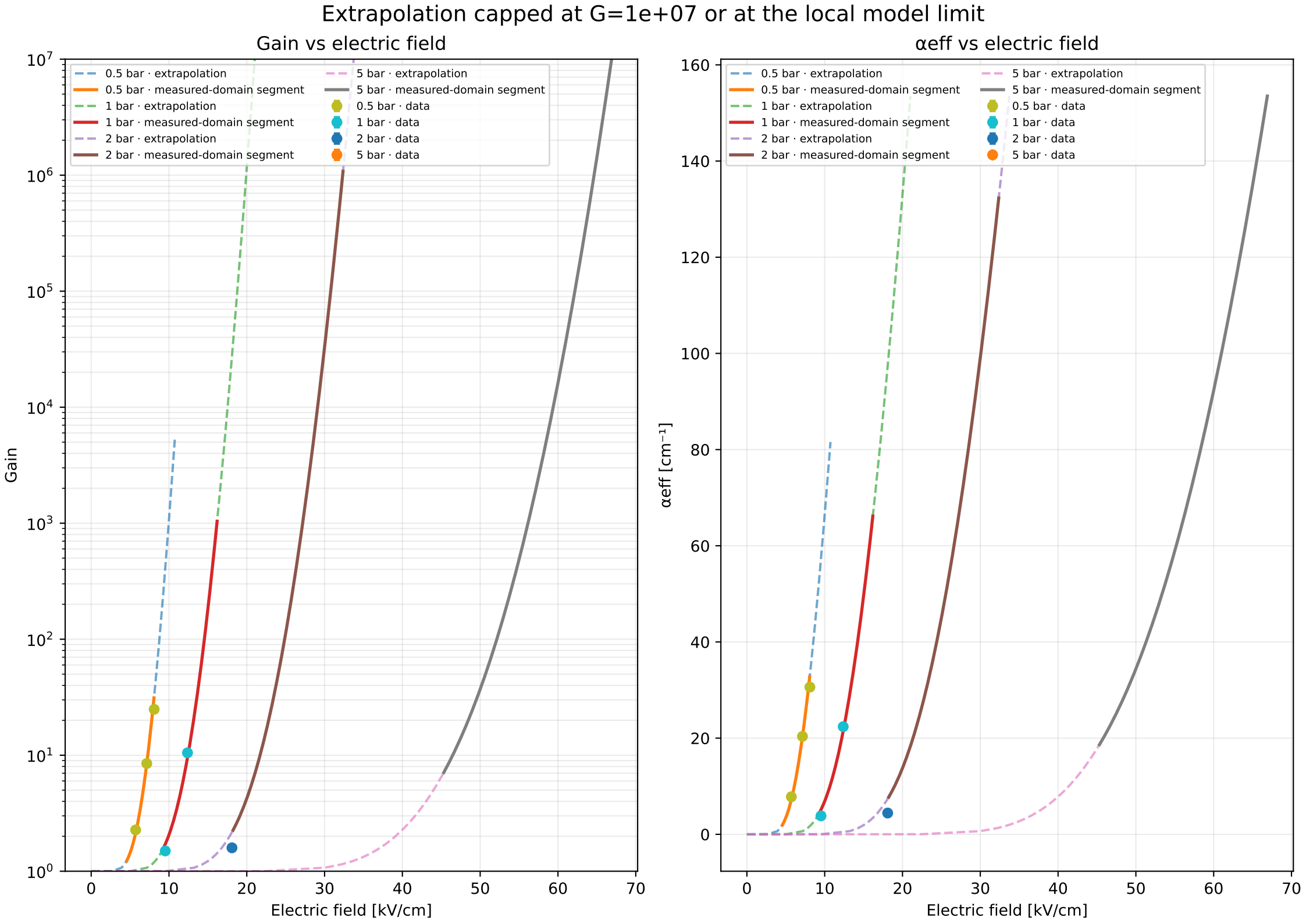
Effective Townsend coefficient vs electric field



Gain-space residuals (±ln 1.2 guides)




```
ROOT runs: 36
Unique physical points: 12
Pressures: 0.5, 1, 2, 5 bar
Selected model: Townsend 2p
Status: rejected_large_cross_validation_outlier
Training RMSE: ×1.276
Cross-validation median: ×1.288
Cross-validation worst: ×3.192
Reduced-field range: 9.048–16.19 kV cm⁻¹ bar⁻¹
A=2568.8, B=59.229, m=0, n=1
Covariance condition: 3.61e+03
Warnings: large_cross_validation_outlier
0.5 bar display limit: G=5.25e+03 (local inversion lim
1 bar display limit: G=1e+07 (requested cap)
2 bar display limit: G=1e+07 (requested cap)
5 bar display limit: G=1e+07 (requested cap)
```


Extrapolation capped at G=1e+07 or at the local model limit
Gain vs electric field
αeff vs electric field
0.5 bar · extrapolation
0.5 bar · measured-domain segment
1 bar · extrapolation
1 bar · measured-domain segment
2 bar · extrapolation
2 bar · measured-domain segment
5 bar · extrapolation
5 bar · measured-domain segment
0.5 bar · data
1 bar · data
2 bar · data
5 bar · data
Gain
αeff [cm⁻¹]
Electric field [kV/cm]
Electric field [kV/cm]

Exact combined points used by the fit · page 1/1

Repeated ROOTs at the same field are pooled once; all source ROOTs remain on disk.

p [bar]	E [kV/cm]	E/p	Gain	σGain	αeff [cm⁻¹]	npe	runs
0.5	5.71428	11.4286	2.27	0.049	7.80743	1600	3
0.5	7.14286	14.2857	8.48625	0.201	20.3662	1600	3
0.5	8.09524	16.1905	24.87438	0.58	30.608	1600	3
1	7.61905	7.61905	0.758125	0.0176	—	1600	3
1	9.52381	9.52381	1.49625	0.0385	3.83773	1600	3
1	12.381	12.381	10.495	0.272	22.3895	1600	3
2	9.52381	4.76191	0.190625	0.00994	—	1600	3
2	15.2381	7.61905	0.588125	0.022	—	1600	3
2	18.0952	9.04762	1.591875	0.0533	4.42774	1600	3
5	9.52382	1.90476	0.03125	0.00435	—	1600	3
5	19.0476	3.80952	0.005333333	0.00188	—	1500	2
5	27.619	5.52381	0.033125	0.00481	—	1600	3

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-8.93	×1.29	×3.19	no	large cross validation outlier